MergeRec: Model Merging for Data-Isolated Cross-Domain Sequential Recommendation

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Abstract

Modern recommender systems trained on domain-specific data often struggle to generalize across multiple domains. Cross-domain sequential recommendation has emerged as a promising research direction to address this challenge; however, existing approaches face fundamental limitations, such as reliance on overlapping users or items across domains, or unrealistic assumptions that ignore privacy constraints. In this work, we propose a new framework, ***MergeRec***, based on *model merging* under a new and realistic problem setting termed *data-isolated cross-domain sequential recommendation*, where raw user interaction data cannot be shared across domains. MergeRec consists of three key components: (1) *merging initialization*, (2) *pseudo-user data construction*, and (3) *collaborative merging optimization*. First, we initialize a merged model using training-free merging techniques. Next, we construct pseudo-user data by treating each item as a virtual sequence in each domain, enabling the synthesis of meaningful training samples without relying on real user interactions. Finally, we optimize domain-specific merging weights through a joint objective that combines a *recommendation loss*, which encourages the merged model to identify relevant items, and a *distillation loss*, which transfers collaborative filtering signals from the fine-tuned source models. Extensive experiments demonstrate that MergeRec not only preserves the strengths of the original models but also significantly enhances generalizability to unseen domains. Compared to conventional model merging methods, MergeRec consistently achieves superior performance, with average improvements of up to 17.21% in Recall@10, highlighting the potential of model merging as a scalable and effective approach for building universal recommender systems. The source code is available at `github.com/DIALLab-SKKU/MergeRec`.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Cross-domain sequential recommendation; model merging; data isolation; task vector

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1 Introduction

Sequential recommendation (SR) aims to predict the next items a user is likely to prefer based on their interaction history. Recent neural SR models [6, 12, 14, 15, 33] employ various architectures to effectively capture sequential dependencies among items. However, they still face inherent challenges such as the cold-start and data sparsity problems [22–24, 29], which limit their generalizability and overall performance.

Cross-domain sequential recommendation (CDSR) has emerged as a promising research direction [3]. CDSR aims to improve recommendation accuracy by either jointly training models across multiple domains [26, 27, 37, 52] or by transferring knowledge from data-rich domains to sparser ones [1, 16, 17, 20]. However, existing CDSR works face three fundamental limitations. (1) **User/Item overlap dependency**: knowledge transfer typically relies on the presence of overlapping users or items across domains. However, such overlap is extremely limited in practice. We observe that only 16 users and 0 items are shared across eight Amazon domains, reflecting the real-world nature of independently operated domains; (2) **Data isolation**: in real-world scenarios, access to raw user data is often restricted due to organizational boundaries or privacy regulations [41, 47, 48]. User logs contain sensitive information and cannot be shared across domains due to privacy restrictions, making domain-specific training data inaccessible; (3) **Low scalability**: joint training across multiple domains incurs substantial computational overhead, making it impractical for large-scale deployment. Consequently, most prior work has been limited to integration of only two or three domains, leaving scalable multi-domain integration largely unresolved.

We suggest that *model merging* [4, 5, 8–11, 18, 25, 31, 32, 38, 40, 43–46, 49, 50] offers an effective alternative for building universal recommender systems. Model merging integrates fine-tuned parameters from multiple domain- or task-specific models into a single unified model. This paradigm provides several advantages that directly address the key limitations of CDSR: (1) It eliminates the need for overlapping users or items across domains; (2) It naturally preserves user privacy, as only model parameters, not sensitive interaction data, are required; (3) It achieves high scalability by avoiding the computational burden of cross-domain joint training.



Figure 1: Illustration of training and test user sequences in sequential recommendation across multiple domains. Each test user sequence (black box) contains all previous interactions, including those from the training period (blue box), highlighting that test data are a superset of training data in real-world scenarios.

In this paper, we explore the feasibility of applying model merging to CDSR under a new, realistic problem setting termed *data-isolated CDSR*. This setting is motivated by practical real-world constraints, where user interaction data can be used only to train domain-specific models and cannot be shared across domains or accessed afterward. Unlike conventional CDSR, which often relies on strong and impractical assumptions (*i.e.*, overlapping users or items), data-isolated CDSR allows domains to be disjoint. Moreover, while privacy-preserving CDSR typically requires access to domain-specific interaction data during model optimization, data-isolated CDSR constructs a universal cross-domain recommender system without accessing any user interaction data, thereby providing a stronger guarantee of user privacy.

Under this setting, however, directly applying existing model merging methods is non-trivial for two key reasons. First, since interaction data are not shared across domains, test-time adaptation schemes, commonly used in the model merging paradigm to optimize merging weights, cannot be applied. Second, even if test data were accessible, leveraging test sequences in sequential recommendation would violate the core assumptions of model merging. While the model merging paradigm explicitly prohibits using training data, these assumptions do not hold in sequential recommender systems. In such systems, test sequences are not independent of the training data but are generated from the same evolving user behavior. Thus, using test-time user interaction sequences during the merging process would inevitably expose training information (Figure 1). Consequently, leveraging test data for model merging is fundamentally incompatible with the data-isolated CDSR setting.

To this end, we propose ***MergeRec***, a novel framework tailored for data-isolated CDSR. MergeRec comprises three key components: (1) *merging initialization*, (2) *pseudo-user data construction*, and (3) *collaborative merging optimization*. First, we synthesize an initial merged model using training-free merging methods based on *task vectors*, defined as the parameter difference between a fine-tuned model and its corresponding pre-trained model, to capture domain-specific knowledge [9]. Next, we construct pseudo-user data by treating each item in every domain as an individual sequence. Despite its simplicity, MergeRec enables the construction of meaningful samples for merging domains without relying on real user data, effectively simulating cold-start users across domains. Finally, we refine domain-specific merging weights through a recommendation-oriented merging objective.

To design an effective objective function for merging recommender systems, we argue that an ideal merged model should satisfy two fundamental requirements. First, it should be able to decode users' multiple intents, which are often reflected in domain-specific sequential patterns. Second, the unified model should exhibit strong ranking ability, accurately prioritizing items with the highest click probability within each domain context. We point out that existing adaptive merging methods, *i.e.*, *AdaMerging*, address only the latter aspect and are therefore insufficient for merging recommender systems (Section 3).

To overcome this limitation, we propose a joint objective that combines: (1) a *distillation* loss, which leverages the prediction distributions of fine-tuned models as soft labels, and (2) a *recommendation* loss, which treats the top-1 predicted item from each fine-tuned model for a pseudo-user in its corresponding domain as a hard label. The distillation loss transfers collaborative filtering (CF) [28] signals from the fine-tuned models to the merged model, and the recommendation loss guides the merged model to accurately rank items according to their likelihood of being clicked.

Extensive experiments demonstrate that MergeRec not only preserves the strengths of the individual source models but also generalizes effectively to unseen domains. Compared with existing merging methods and strong baselines, including fine-tuned and joint learning models, MergeRec consistently achieves superior performance. Specifically, MergeRec outperforms joint learning and AdaMerging by average gains of 8.72% and 17.21% on Recall@10, respectively. These results highlight that model merging can be a scalable and efficient paradigm for building universal recommender systems.

Our contribution can be summarized as follows:

- **Thorough empirical analysis**: We provide empirical evidence demonstrating that entropy-based optimization, though effective in computer vision and natural language processing, fundamentally fails to capture the multi-intent behavioral patterns inherent in recommender systems.
- **The first model merging framework for recommender systems**: We propose ***MergeRec***, a task vector-based model merging framework tailored for recommender systems. MergeRec comprises three key components: (1) training-free merging initialization, (2) privacy-preserving pseudo-user data construction, and (3) a recommendation-oriented merging objective.
- **Comprehensive evaluation**: Through extensive experiments across eight Amazon benchmark datasets and four backbone architectures, we demonstrate that MergeRec consistently outperforms existing model merging baselines. Notably, MergeRec exhibits superior generalizability to unseen domains and robust performance under data-scarce conditions.

2 Preliminaries

2.1 Cross-domain Sequential Recommendation

Let $\mathcal{D} = \{D_1, D_2, ..., D_K\}$ denote the set of all recommendation domains, where D_k denotes the k-th domain. Each domain D_k consists of a set of items $\mathcal{I}_k$ and users $\mathcal{U}_k$. For an arbitrary user $u \in \mathcal{U}_k$, the interaction history is represented as an ordered sequence of items based on timestamps: $u = \left[i_1, i_2, \ldots, i_{|u|}\right]$, where $|u|$ denotes the number of interactions of user u. CDSR models aim to predict and rank items in $\mathcal{I}_k$ by estimating the probability that user u will interact with each item next, conditioned on the user's past interactions:

$$\theta^* = \arg\max_{\theta} P\big(i = i_{|u|+1} \mid u, \theta\big), \tag{1}$$

where θ denotes the parameters of the CDSR model.

2.2 Text-based Sequential Recommendation

Text-based SR [7, 13] leverages pre-trained language models (PLMs) to encode item-level textual information. By representing both users and items through textual descriptions, this approach enables recommendations for previously unseen (*i.e.*, cold-start) items.

Formally, the textual representation of an item t_i is constructed from its attribute descriptions (*e.g.*, title, brand, and category). The textual representation of a user t_u is defined as the concatenation of the textual representations of all items the user has interacted with:

$$t_u = \left[t_{i_1}; t_{i_2}; \ldots; t_{i_{|u|}}\right], \tag{2}$$

where ; denotes the concatenation operator.

Let $f(\cdot \mid \theta_{\mathbf{k}})$ denote a PLM-based encoder with parameters $\theta_k \in \mathbb{R}^P$ fine-tuned on domain D_k, where P is the total number of model parameters. Given the textual inputs t_u and t_i, the encoder produces a user representation vector $\mathbf{r}_u \in \mathbb{R}^d$ and an item representation vector $\mathbf{r}_i \in \mathbb{R}^d$ by extracting the final hidden state representations:

$$\mathbf{r}_u = f\left(t_u \mid \theta_k\right), \ \mathbf{r}_i = f\left(t_i \mid \theta_k\right), \tag{3}$$

where d denotes the dimension of the final hidden representations.

The recommendation score $\hat{y}_{ui}$ between user u and item i is computed as the cosine similarity between their representation vectors:

$$\hat{y}_{ui} = \cos(\mathbf{r}_u, \mathbf{r}_i). \tag{4}$$

The model parameters θ_k are optimized using a cross-entropy objective:

$$\theta_k^* = \arg\min_{\theta_k} \sum_{u \in \mathcal{U}_k} \left(\log \hat{y}_{k,ui^+} + \sum_{i^- \in \mathcal{I}_k} \log\left(1 - \hat{y}_{k,ui^-}\right) \right), \tag{5}$$

where i^+ denotes the next item in the user sequence, and i^- represents negative items in the domain $\mathcal{I}_k$ excluding i^+. Note that the domain-specific parameters θ_k are initialized from the pre-trained base model parameters $\theta_{base} \in \mathbb{R}^P$.

3 Proposed Method: MergeRec

As illustrated in Figure 2, we design the ***MergeRec*** framework to address the practical constraint that no interaction data can be shared across domains, termed *data-isolated CDSR*. MergeRec consists of three key components: (1) *Merging Initialization*, which consolidates multiple domain-specific fine-tuned models into a single unified model that integrates knowledge across domains; (2) *Pseudo-user Data Construction*, which synthesizes meaningful merging samples without relying on real user interactions; and (3) *Collaborative Merging Optimization*, which jointly optimizes a recommendation loss and a knowledge distillation loss to enable recommendation-aware parameter integration. Through this design, MergeRec effectively preserves domain-specific CF signals while ensuring strong generalizability across multiple domains.

Table 1: Categorization of cross-domain sequential recommendation problem settings.

Setting	No User/Item Overlap Required	Privacy-Aware	Data-Isolated
Conventional CDSR [1, 3, 16, 17, 27]	✗	✗	✗
Privacy-preserving CDSR [19, 34, 39, 42, 48]	✗	▲	✗
Data-isolated CDSR (Proposed)	✓	✓	✓

3.1 Data-isolated CDSR

We formalize a new and realistic setting for cross-domain sequential recommendation, termed *data-isolated CDSR*. As shown in Table 1, this setting is characterized by two key requirements: (i) it assumes no overlap across domains, (ii) it prohibits access to user interaction data during cross-domain model construction, in contrast to conventional CDSR. Under data-isolated CDSR, domains may be entirely disjoint in both users and items (*e.g.*, $\mathcal{U}_k \cap \mathcal{U}_{k'} = \emptyset$ and $\mathcal{I}_k \cap \mathcal{I}_{k'} = \emptyset$ for $k \neq k'$), reflecting real-world environments that are independently operated. Moreover, this setting enforces a strict *data isolation* constraint: raw interaction logs are accessible only within each domain for training domain-specific models and cannot be shared across domains. Consequently, a cross-domain recommender system must be constructed *without* accessing any domain-specific interaction data. Instead, we assume access only to K domain-specific fine-tuned models $\{\theta_k\}_{k=1}^K$, while the datasets used to train these models remain completely inaccessible. The goal of data-isolated CDSR is to produce a single *universal* sequential recommender system that can operate across multiple domains.

3.2 Merging Initialization

Problem definition. Let $f_{\theta_k}(u_k) \rightarrow \hat{y}_k$ denote an SR model fine-tuned on the private data of domain $D_k = \{\mathcal{U}_k, \mathcal{I}_k\}$. For an arbitrary user interaction sequence $u_k \in \mathcal{U}_k$, the model outputs a click probability vector $\hat{\mathbf{y}}_k \in \mathbb{R}^{|\mathcal{I}_k|}$ over candidate items. Without loss of generality, we assume that the model parameters are composed of L layers, *i.e.*, $\theta = \left\{\theta^1, \theta^2, ..., \theta^L\right\}$.

Task vector. A task vector represents the parameter shift required to adapt a pre-trained model to a specific downstream task [4, 8, 9, 11, 25, 31, 38, 44, 46, 49, 50]. In our context, each task corresponds to *recommendation within a specific domain*. Accordingly, the task vector captures domain-specific knowledge, enabling a pre-trained model to specialize in that domain.

Formally, the task vector $\boldsymbol{\tau}_k \in \mathbb{R}^P$ for domain k is defined as the difference between the parameters of the fine-tuned model θ_k and

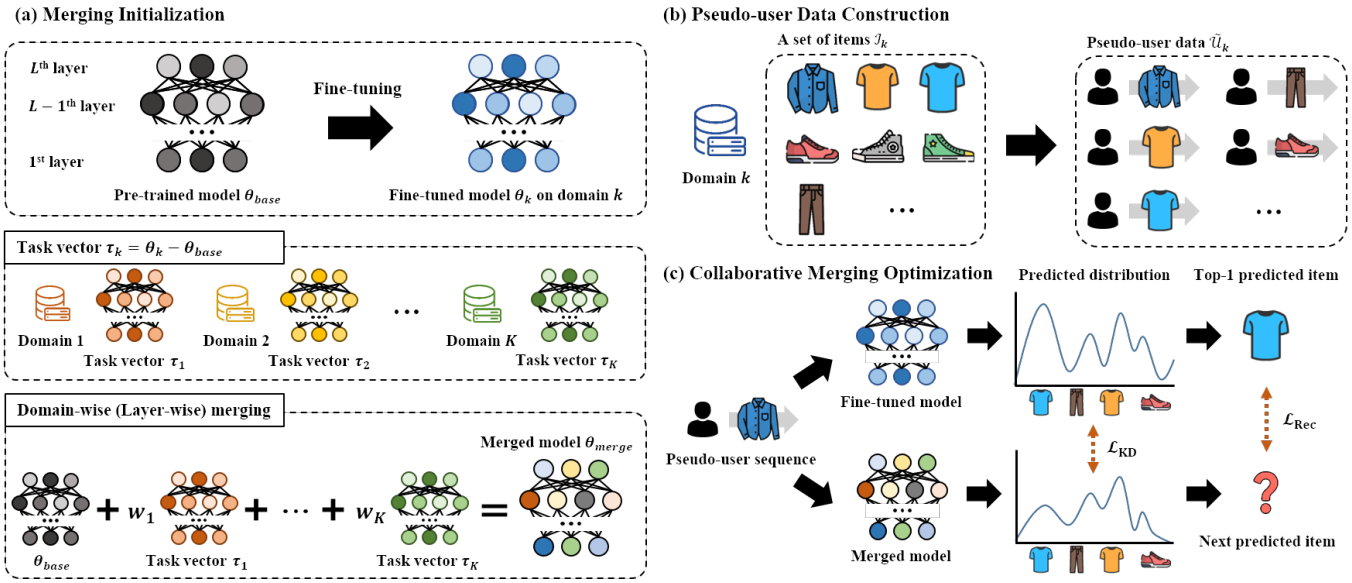


Figure 2: Overview of MergeRec with three main components. (a) Merging initialization integrates into a unified model containing multi-domain knowledge. (b) Pseudo-user data construction creates a single-item sequence. (c) Collaborative merging optimization jointly optimizes the recommendation loss $\mathcal{L}_{Rec}$ and the distillation loss $\mathcal{L}_{KD}$.

those of the original pre-trained base model θ_{base}:

$$\boldsymbol{\tau}_k = \theta_k - \theta_{base}, \tag{6}$$

where P denotes the total number of model parameters.

Domain-wise merging. Domain-wise merging integrates multiple fine-tuned models by combining their task vectors, each weighted by a domain-specific scalar w_k, and adding them to the base model parameters. The merging weights $\mathbf{w} = \{w_1, ..., w_K\}$ can be either uniformly assigned [9, 44] or adaptively learned to reflect domain characteristics [46]. Intuitively, domains containing more distinctive knowledge may receive higher weights, while those sharing similar CF signals may be down-weighted. Formally, domain-wise merging is defined as:

$$\theta_{merge} = \theta_{base} + \sum_{k=1}^{K} w_k \cdot \boldsymbol{\tau}_k. \tag{7}$$

In this work, we learn the domain-specific weights $\mathbf{w}$ in a data-driven manner.

Layer-wise merging. Since different layers in deep neural networks capture different levels of abstraction [30, 36], applying a single scalar weight per domain may be insufficient to control inter-domain interference. To enable fine-grained integration, we assign independent merging weights to each layer for every domain.

Let $\theta_k = \{\theta_k^1, \ldots, \theta_k^L\}$ denote the parameters of the fine-tuned model for domain k. The corresponding layer-wise task vector is defined as $\boldsymbol{\tau}_k = \{\theta_k^1 - \theta_{\text{base}}^1, ..., \theta_k^L - \theta_{\text{base}}^L\}$. The layer-wise merging is then defined as:

$$\theta_{merge} = \left\{ \theta_{base}^l + \sum_{k=1}^{K} w_k^l \cdot \tau_k^l \right\}_{l=1}^{L}. \tag{8}$$

The layer-specific merging weights $\mathbf{w}_k = \{w_k^1, \ldots, w_k^L\}$ are similarly learned in a data-driven manner.

3.3 Pseudo-user Data Construction

User logs in recommender systems typically contain sensitive personal information and cannot be shared across domains, making it challenging to construct data for learning merging weights. To address this, we propose a novel pseudo-user data construction strategy that represents each item in a domain as a single-item interaction sequence. Our design is grounded in the idea that CF knowledge is encapsulated within domain-specific models and can be transferred without relying on the real user data on which they were trained. By leveraging pseudo-users as surrogate inputs, our approach enables learning merging weights without access to domain-specific fine-tuning data while strictly preserving data isolation.

Formally, we construct the pseudo-user set for domain k as:

$$\tilde{\mathcal{U}}_k = \{[i] \mid i \in \mathcal{I}_k\}. \tag{9}$$

These synthesized samples emulate plausible cold-start users in each domain and thus provide meaningful signals for model merging. Although each pseudo-user sequence contains no explicit sequential context, it serves as a probe to elicit rich CF knowledge encoded in the corresponding domain-specific model f_{θ_k}. We therefore employ each domain-specific model as a teacher, whose conditional distribution $P_{\theta_k}(\cdot \mid [i])$ captures the local co-consumption structure around item i, *i.e.*, next-item likelihoods. By distilling the merged model to align with these teacher distributions, we effectively transfer domain-specific CF signals. We observe that even single-item pseudo-user sequences are sufficient for effective model merging (Section 5), providing a practical and privacy-preserving foundation for collaborative merging optimization. While extending pseudo-user sequences to longer contexts may further enrich the transferred signals, we leave this promising direction for future work.

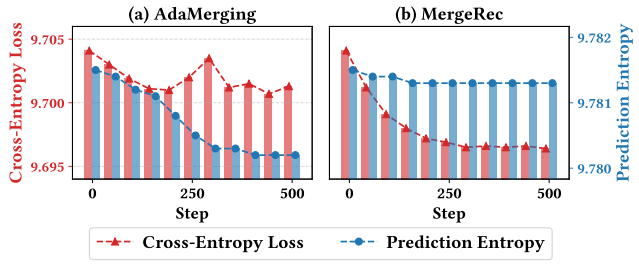


Figure 3: Cross-entropy loss and prediction entropy dynamics of AdaMerging and MergeRec (Ours) over training steps.

3.4 Collaborative Merging Optimization

Inadequacy of entropy-based optimization. We examine a representative adaptive merging method, AdaMerging [46], from the perspective of recommender systems. Figure 3 illustrates the training dynamics of cross-entropy loss and prediction entropy for both AdaMerging and our proposed ***MergeRec*** over training iterations on eight datasets, evaluated using test user sequences. The cross-entropy loss reflects alignment with the recommendation objective, whereas prediction entropy measures the confidence of the merged model's predictions. As shown in Figure 3 (a), AdaMerging successfully reduces prediction entropy during training but fails to achieve a corresponding decrease in cross-entropy loss. This limitation stems from its exclusive focus on entropy minimization, which merely amplifies confidence in the top-1 predicted item. However, users often exhibit *multi-intent* behavioral patterns rather than a single dominant intent [2, 35, 51], making entropy-based optimization alone insufficient to capture the rich CF signals learned by domain-specific fine-tuned models.

To overcome this, we introduce a distillation loss that extends beyond entropy-based optimization by explicitly aligning the merged model with teacher distributions derived from domain-specific models. MergeRec (Figure 3 (b)) simultaneously reduces both cross-entropy loss and prediction entropy, demonstrating more consistent and effective optimization toward the recommendation objective.

Joint objective function. We posit that an ideal merged model for cross-domain recommendation should simultaneously satisfy two essential aspects. First, it should effectively capture diverse user intents reflected in behavioral patterns within each domain and retrieve items relevant to those intents. This requires successfully transferring domain-specific CF knowledge from fine-tuned models to the merged model. Second, the merged model should exhibit strong discriminative capability to accurately identify the items that users are most likely to click on within each domain. To jointly address these requirements, we propose the following optimization function:

$$\mathcal{L} = \mathcal{L}_{\text{Rec}} + \lambda \cdot \mathcal{L}_{\text{KD}}, \tag{10}$$

where λ is a hyperparameter that balances the two losses. In Eqs. (7) and (8), we optimize only the merging weights w, while keeping the base model parameters θ_{base} and the task vector τ fixed. Since only K or $K \times L$ domain-specific weights are optimized, MergeRec provides a computationally efficient solution.

The knowledge distillation loss $\mathcal{L}_{\text{KD}}$ integrates domain-specific CF knowledge encoded in fine-tuned models by aligning the predictions of the merged model with those of the corresponding domain-specific models. For a pseudo-user sequence $u \in \tilde{\mathcal{U}}_k$ in domain k, we minimize the Kullback-Leibler (KL) divergence between the prediction $\hat{y}_{merge}$ produced by the merged model θ_{merge} and the prediction $\hat{y}_k$ produced by the fine-tuned model θ_k:

$$\begin{aligned}\mathcal{L}_{\text{KD}} &= \sum_{k=1}^{K} \sum_{u \in \tilde{\mathcal{U}}_k} \text{KL}(\hat{p}_{merge,u} \parallel \hat{p}_{k,u}) \\ &= \sum_{k=1}^{K} \sum_{u \in \tilde{\mathcal{U}}_k} \sum_{i \in \mathcal{I}_k} \hat{p}_{merge,ui} \log \frac{\hat{p}_{merge,ui}}{\hat{p}_{k,ui}},\end{aligned} \tag{11}$$

where $\hat{p}_* = \text{softmax}(\hat{y}_*/T)$, and T denotes the temperature hyperparameter which is empirically set to 1 in our experiments.

The recommendation loss $\mathcal{L}_{\text{Rec}}$ encourages the merged model to accurately identify items aligned with user intent by assigning high scores to the next item and low scores to others. Since real user sequences are unavailable, we leverage the top-1 predicted item $\tilde{i}^+$, obtained by feeding pseudo-user data into the corresponding fine-tuned model, as a positive pseudo-label:

$$\mathcal{L}_{\text{Rec}} = \sum_{k=1}^{K} \sum_{u \in \tilde{\mathcal{U}}_k} \left(\log \hat{y}_{merge,u\tilde{i}^+} + \sum_{i^- \in \mathcal{I}_k} \log\left(1 - \hat{y}_{merge,ui^-}\right) \right). \tag{12}$$

4 Experimental Setup

Datasets. To simulate a cross-domain recommendation environment, we use eight categories from the Amazon dataset[1] [2]: Arts, Beauty, Instruments, Office, Pantry, Scientific, Sports, and Toys. Following existing work [12, 33], we adopt a 5-core setting, *i.e.*, users and items with fewer than five interactions are removed. Detailed dataset statistics are provided in Appendix A.

Baselines. We compare MergeRec with the following methods:

- **Zero-shot**: Directly applies pre-trained text-based SR models without fine-tuning on a specific domain.
- **Fine-tuning**: Fine-tunes pre-trained models using domain-specific interaction data.
- **Joint Learning**: Trains a unified model on aggregated multi-domain datasets with shared parameters.
- **Task Arithmetic** [9]: Constructs a cross-domain model by linearly adding task vectors to a pre-trained model.
- **TIES** [44]: Reduces noise and conflicts between task vectors by selecting parameters with large variance and aligning their signs.
- **AdaMerging** [46]: Learns adaptive merging weights in an unsupervised manner by minimizing the prediction entropy of the merged model.

We evaluate all methods on RecFormer-base/large [13], a representative text-based SR model, and BLaIR-base/large [7], a language model post-trained on a recommendation corpus, as backbone architectures. For a fair comparison under the data-isolation setting, AdaMerging is adapted to use the same pseudo-user data as MergeRec, treating it as unlabeled inputs for entropy-based optimization. Implementation details are provided in Appendix B.

[1] https://cseweb.ucsd.edu/~jmcauley/datasets/amazon/links.html
[2] https://cseweb.ucsd.edu/~jmcauley/datasets/amazon_v2/

Table 2: Performance comparison with six baseline methods on four backbone models, *i.e.*, RecFormer-base/large [13] and BLaIR-base/large [7]. We report normalized Recall@10 performance (%) relative to the fine-tuned model's performance, which is 100%. The best results are marked in bold, and the second-best results are shown as underlined. '*' indicates the statistically significant gain of MergeRec over the best baseline model ($p < 0.02$ for one-tailed t-test).

Backbone	Method	Avg.	Arts	Beauty	Inst.	Office	Pantry	Sci.	Sports	Toys
RecFormer-base	Zero-shot	75.46	76.04	63.66	72.15	61.61	74.23	92.85	71.11	84.94
	Joint Learning	80.17	77.37	79.53	83.26	73.37	85.34	79.96	69.62	92.94
	Weight Averaging	89.84	91.20	83.05	90.86	77.39	93.62	99.68	**86.66**	93.00
	Task Arithmetic	88.95	91.52	81.45	88.12	83.79	92.67	98.90	81.32	82.34
	TIES	91.08	93.29	**85.60**	90.57	**88.22**	92.75	97.68	85.56	86.25
	AdaMerging (Domain-wise)	78.91	87.26	58.66	75.81	64.78	75.02	92.56	67.97	93.11
	AdaMerging (Layer-wise)	67.36	67.17	32.92	67.36	58.54	65.64	84.54	65.24	84.93
	MergeRec (Domain-wise)	**92.33***	**96.14***	83.69	90.93	84.27	**95.56***	100.73*	84.00	**93.35***
	MergeRec (Layer-wise)	92.08*	95.45*	84.85	**91.13***	83.01	94.86*	**101.44***	85.07	92.14
RecFormer-large	Zero-shot	59.07	72.57	51.89	46.03	39.23	44.34	82.00	62.54	63.57
	Joint Learning	83.73	83.14	79.48	79.76	73.61	90.27	92.99	91.49	83.61
	Weight Averaging	91.23	92.27	87.10	88.94	79.13	**96.46**	**98.32**	**98.88**	94.36
	Task Arithmetic	87.99	91.32	83.32	88.26	82.78	92.08	92.99	87.80	81.14
	TIES	89.96	93.00	85.74	90.25	**88.17**	92.91	94.10	89.09	80.79
	AdaMerging (Domain-wise)	72.59	83.37	59.49	77.55	63.74	61.37	83.96	58.14	70.69
	AdaMerging (Layer-wise)	70.80	79.59	52.28	68.27	60.64	67.12	83.08	72.97	72.01
	MergeRec (Domain-wise)	**92.99***	**95.19***	**90.08***	**91.75***	83.39	96.20	97.41	94.74	**96.64***
	MergeRec (Layer-wise)	92.50*	94.21*	89.77*	90.25	82.18	96.27	97.86	96.53	96.17*
BLaIR-base	Zero-shot	41.10	45.88	35.57	31.55	27.74	47.20	54.94	34.50	42.32
	Joint Learning	83.67	82.89	**97.30**	**83.52**	73.09	91.24	74.85	**92.88**	**94.87**
	Weight Averaging	87.90	91.98	93.97	78.41	76.12	**91.39**	**99.95**	89.45	81.81
	Task Arithmetic	61.50	57.05	53.43	66.66	61.61	62.42	73.27	49.34	53.78
	TIES	82.95	84.37	78.02	78.26	**87.64**	77.18	88.52	87.26	77.39
	AdaMerging (Domain-wise)	60.60	67.93	52.07	53.40	44.83	50.80	72.94	72.91	73.14
	AdaMerging (Layer-wise)	68.48	73.09	61.14	68.86	55.34	57.54	83.92	74.23	70.61
	MergeRec (Domain-wise)	87.40	**94.42***	89.38	77.75	81.44	85.18	95.48	85.05	84.01
	MergeRec (Layer-wise)	**88.01**	93.53*	91.83	78.98	81.15	87.82	96.55	87.45	83.06
BLaIR-large	Zero-shot	33.44	38.42	29.99	24.55	19.23	46.48	37.83	27.84	41.70
	Joint Learning	84.59	83.33	81.01	86.78	78.56	90.60	78.48	83.85	**100.63**
	Weight Averaging	88.99	92.76	89.13	83.82	76.25	93.20	103.56	86.36	81.69
	Task Arithmetic	78.86	82.64	79.11	65.86	74.43	83.64	93.56	73.13	67.87
	TIES	90.90	93.78	91.09	87.76	**90.01**	92.58	98.36	**93.67**	75.35
	AdaMerging (Domain-wise)	69.27	82.21	62.36	71.76	51.16	60.90	78.41	77.27	68.45
	AdaMerging (Layer-wise)	78.22	84.78	73.80	78.57	62.43	83.01	84.04	71.51	83.00
	MergeRec (Domain-wise)	91.80*	97.80*	93.07*	**91.81***	80.51	89.76	102.88	90.16	82.84
	MergeRec (Layer-wise)	**93.70***	**98.73***	**95.40***	91.74*	82.62	**94.17***	**105.18***	90.28	85.63

Evaluation protocol. Following [12, 33], we adopt the leave-one-out strategy to split the train, validation, and test datasets. For each user, the most recently interacted item is used for testing, the second most recently interacted item for validation, and the rest for training. Note that the training data is used only for fine-tuning and joint learning. We evaluate recommendation performance using Recall@10 (R@10). Following [44], we normalize the performance of each method by that of its corresponding fine-tuned model. The normalized results are reported in Table 2 and Figures 4, 5, and 7.

5 Experimental Results

5.1 Overall Performance

Table 2 shows the normalized R@10 of MergeRec and seven baseline methods across eight datasets and four backbone models, where the performance of the fine-tuned model on each dataset is normalized to 100%. The corresponding unnormalized R@10 results are provided in Appendix C.

MergeRec consistently achieves the best average performance across all datasets and backbone models for both the domain-wise and layer-wise variants. Specifically, MergeRec outperforms Joint

Table 3: Performance comparison over varying the data sparsity of a target-domain training set. We merge five source models (trained on Arts, Beauty, Pantry, Sports, and Toys) with one target model for each of the three datasets (Inst., Office, and Sci.). Each target model is trained on a subset of the full dataset (1%, 5%, and 10%). 'Ratio' denotes the fraction of target-domain training data used for fine-tuning, and the RecFormer-base is used as the backbone. The metric is Recall@10.

Ratio	Method	Avg.	Inst.	Office	Sci.
1%	Fine-tuning	0.0989	0.0745	0.0953	0.1268
	Task Arithmetic	0.1069	0.0828	0.0995	0.1385
	TIES	0.0650	0.0158	0.0614	0.1178
	AdaMerging	0.0982	0.0732	0.0898	0.1318
	MergeRec	**0.1089**	**0.0859**	**0.1013**	**0.1394**
5%	Fine-tuning	0.1057	0.0752	0.1094	0.1325
	Task Arithmetic	0.1101	0.0859	0.1050	0.1392
	TIES	0.0800	0.0812	0.1017	0.0570
	AdaMerging	0.1047	0.0814	0.1023	0.1304
	MergeRec	**0.1128**	**0.0888**	**0.1098**	**0.1399**
10%	Fine-tuning	0.1118	0.0838	**0.1175**	0.1341
	Task Arithmetic	0.1107	0.0860	0.1055	**0.1405**
	TIES	0.0840	0.0809	0.1017	0.0693
	AdaMerging	0.0982	0.0755	0.0935	0.1258
	MergeRec	**0.1120**	**0.0887**	0.1088	0.1386

Learning and AdaMerging with average gains of 8.72% and 17.21%, respectively. This indicates that MergeRec simultaneously enhances the ranking discriminative ability of the merged model and effectively transfers domain-specific CF knowledge from fine-tuned models. Meanwhile, AdaMerging performs substantially worse across all datasets and backbone models, suggesting that merely amplifying prediction confidence is insufficient to capture the diverse CF signals present across multiple domains. Furthermore, MergeRec surpasses training-free model merging methods (*i.e.*, Task Arithmetic and TIES) by average gains of 9.90% and 3.04%, respectively.

Several model merging methods, *i.e.*, MergeRec, Weight Averaging, and TIES, consistently outperform Joint Learning. These results demonstrate that model merging can effectively capture complementary domain knowledge and improve recommendation quality without relying on cross-domain training data. It further highlights the practical advantages of model merging, as it not only reduces computational overhead but also enables synergistic knowledge transfer across domains without end-to-end re-training.

Cross-domain merging is particularly beneficial for data-scarce domains. On the Scientific dataset, MergeRec achieves improvements of 1.44% for RecFormer-base (Domain-wise) and 5.18% for BLaIR-large (Layer-wise) compared to their respective fine-tuned counterparts. These improvements can be attributed to the limited number of users in the Scientific domain, where the merged model benefits more substantially from cross-domain knowledge transferred from other domains.

Overall, these results demonstrate that MergeRec provides a robust and scalable solution for cross-domain model merging in recommender systems, delivering consistent and significant performance gains across diverse domains and backbone architectures.

Table 4: Unseen-domain performance comparison of task-vector based model merging methods using the RecFormer-base backbone. We train a merged model on the Arts, Beauty, Pantry, Sports, and Toys datasets and test it on the Inst., Office, and Sci. datasets. The metric is Recall@10.

Method	Avg.	Inst.	Office	Sci.
Task Arithmetic	0.1062	0.0817	0.0984	0.1385
TIES	0.1062	0.0817	0.0988	0.1380
AdaMerging	0.0998	0.0759	0.0901	0.1333
MergeRec	**0.1081**	**0.0849**	**0.1002**	**0.1393**

5.2 Model Merging on Scarce Training Data

Collecting sufficient data is often challenging in the early stages of recommender systems. Under such data-scarce conditions, model merging can offer a promising solution for improving model generalization by leveraging knowledge from data-rich domains. We investigate whether model merging can improve recommendation performance in domains with limited data.

To simulate this scenario, we divide the eight domains into two groups: five source domains (Arts, Beauty, Pantry, Sports, Toys) and three target domains (Instruments, Office, Scientific). We then vary the degree of scarcity in the target domain by randomly sampling $k\%$ of users ($k = 1, 5, 10$) from the whole user set to construct data-scarce training sets. Each fine-tuned model trained on a data-scarce target domain is subsequently merged with the fine-tuned models trained from the five source domains.

Table 3 compares MergeRec with five merging methods under different levels of data scarcity using the RecFormer-base backbone. MergeRec consistently outperforms the corresponding fine-tuned models, demonstrating its strong ability to transfer knowledge across domains even under extreme data scarcity. Moreover, MergeRec consistently surpasses AdaMerging across all scarcity levels, indicating that our recommendation-oriented optimization captures transferable CF patterns more effectively than the entropy minimization approach. These results confirm that MergeRec can reliably transfer CF signals from data-rich to data-scarce domains.

5.3 Performance on Unseen Domains

To examine whether model merging remains effective under extreme conditions where no interaction data are available for the target domain, we merge models trained on five source domains (Arts, Beauty, Pantry, Sports, Toys) and evaluate their performance on three unseen target domains (Inst., Office, Sci.). Note that neither interaction data nor domain-specific models from the target domains are used in constructing the merged model.

Table 4 shows the performance of four merging methods on unseen domains. MergeRec consistently outperforms the other methods, achieving improvements of up to 3.92% over Task Arithmetic. These results demonstrate the strong generalizability of MergeRec and highlight its effectiveness in transferring knowledge to entirely unseen domains without relying on any target-domain data.

5.4 Further Analysis

Effect of the number of merged domains. To assess the robustness of MergeRec, we analyze how recommendation performance varies with the number of merged models, as shown in Figure 4.

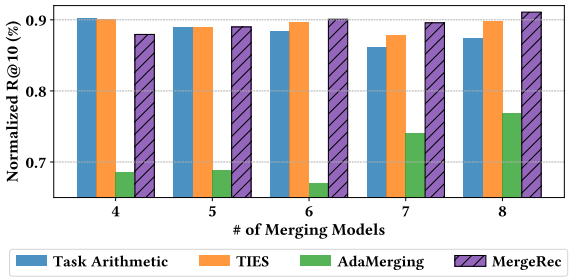


Figure 4: Normalized average performance across varying the number of datasets for model merging.

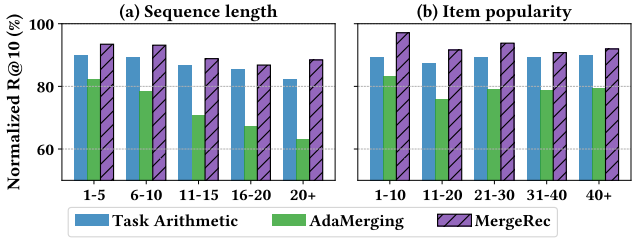


Figure 5: Normalized average performance across user and item groups on eight datasets. User and item groups are divided by sequence length and item popularity, respectively.

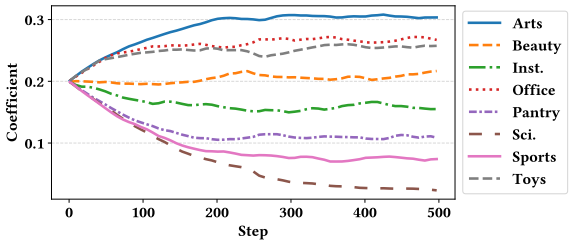


Figure 6: Domain-specific coefficients dynamics over training steps on the RecFormer-base backbone.

Task Arithmetic and TIES exhibit competitive performance when merging a small number of domains (four or five), but their performance deteriorates as more domains are included, eventually falling behind MergeRec. AdaMerging consistently performs poorly across all settings, indicating that its entropy-based optimization strategy fails to effectively adapt to recommendation tasks. In contrast, MergeRec maintains stable performance regardless of the number of merged domains. Notably, its average performance improves as more domains are integrated, highlighting its strong ability to effectively consolidate and leverage knowledge across an increasing number of diverse domains.

Performance across user and item groups. To further analyze the sources of performance improvements, we partition the test data into multiple groups based on user history length and target item popularity. Figure 5 shows the performance of three merging methods across five sequence-length groups and five item popularity ranges. All methods are evaluated on the eight datasets, and the results are averaged and normalized by the performance of the corresponding fine-tuned model within each group.

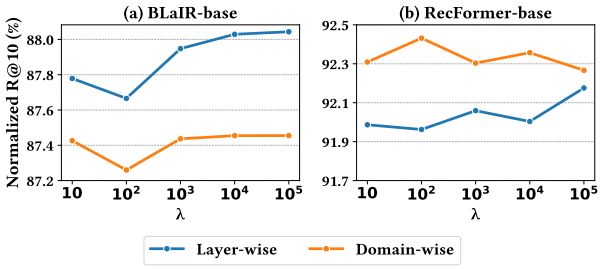


Figure 7: Normalized average performance over varying λ.

As shown in Figure 5 (a), AdaMerging exhibits severe performance degradation as the sequence length increases. This result indicates that AdaMerging struggles to capture CF signals in longer sequences, as it does not learn inter-item relationships during optimization. In contrast, MergeRec consistently achieves the best performance across all sequence lengths by effectively transferring inter-item relationships through the proposed objective function. Figure 5 (b) shows that MergeRec outperforms other methods across all item popularity ranges. Notably, in the least popular item group (1–10), MergeRec achieves substantial performance gains over competing methods, highlighting its ability to leverage cross-domain knowledge to recommend less popular items.

Domain-specific weight dynamics. To understand how the model adapts to different domains during merging, we analyze the training trajectories of the domain-wise merging weights $\mathbf{w}$ across eight domains using the RecFormer-base model. Figure 6 illustrates the evolution of each domain's weight during training. We observe that the weights gradually converge to distinct values for each domain, indicating that the model learns to differentiate the relative importance of individual domains. Notably, domains with larger scales (*e.g.*, Arts and Office) converge to higher weight coefficients. These results suggest that domains exhibiting more complex CF patterns are assigned greater emphasis, enabling the merged model to better preserve domain-specific knowledge. The corresponding results for other backbone models are reported in Appendix D.

5.5 Hyperparameter Sensitivity

Figure 7 depicts the effect of the trade-off hyperparameter λ on recommendation performance. The results are averaged across all eight datasets and normalized by the performance of the corresponding fine-tuned models. We find that performance generally improves as λ increases, except for domain-wise merging with the RecFormer-base backbone. This suggests that placing greater emphasis on the knowledge distillation loss is typically more effective. This can be attributed to the model learning the full item-prediction distribution from domain-specific teacher models, which enables the merged model to capture richer and more informative CF signals. In contrast, methods that focus solely on amplifying top-1 prediction confidence provide a more limited supervisory signal.

6 Related Work

6.1 Cross-Domain Sequential Recommendation

CDSR [3] has emerged as an effective approach to alleviate data sparsity and cold-start problems in single-domain recommender

systems [6, 12, 14, 15, 33]. Existing CDSR approaches can be categorized into three types: single-target CDSR, dual-target CDSR, and multi-target CDSR. Additionally, privacy-preserving cross-domain recommendation (CDR) has gained increasing attention in privacy-sensitive and federated learning scenarios.

Single-target CDSR. Single-target CDSR [1, 16, 17, 20] is the most extensively studied setting, aiming to improve recommendation performance in a data-scarce target domain by transferring knowledge from data-rich source domains with abundant user-item interactions or auxiliary information. C^2DSR [1] jointly models intra-sequence and inter-sequence item relationships through graph neural networks and self-attention mechanisms. To enhance self-attention modules, MAN [16] introduces both local and global attention modules to capture domain-specific and cross-domain information. Recently, LLM4CDSR [17] leverages large language models to generate semantic item representations from textual attributes and hierarchical user profiles from interaction sequences, facilitating cross-domain knowledge transfer.

Dual- and multi-target CDSR. Dual- and multi-target CDSR [26, 27, 37, 52] aims to simultaneously improve recommendation performance across multiple domains. DTCDR [52] adopts a multi-task learning framework that bidirectionally transfers user preference representations across domains via shared embeddings. Recent studies [26, 27] have highlighted the negative transfer problem, where knowledge from other domains can fail to provide beneficial contributions. To address this, CGRec [26] and SyNCRec [27] introduce adaptive loss weighting based on estimated transfer gaps between single-domain and cross-domain sequential recommendation tasks. However, these approaches fundamentally rely on overlapping users or items to enable knowledge transfer and do not account for the privacy constraints commonly required in real-world scenarios.

Privacy-preserving CDR. Privacy-preserving CDR [19, 34, 39, 42, 48] aims to improve recommendation performance under settings where access to raw data from individual domains is restricted or entirely unavailable. An early study [42] separates personalized and transferable components to enable privacy-compliant recommendations. P2M2-CDR [39] disentangles domain-common and domain-specific embeddings while applying local differential privacy to perturb shared representations. FedGCDR [48] further adopts federated graph learning with differential privacy-based knowledge extraction and graph expansion to mitigate the negative transfer problem. However, these approaches typically rely on overlapping users across domains and require cross-domain coordination during training, where domain-specific data are used alongside data from other domains (*e.g.*, exchanged model parameters or gradients). Therefore, they do not fully satisfy the *data isolation* constraint.

In this paper, *we pioneer the application of model merging to data-isolated multi-target CDSR*, providing a scalable solution for integrated multi-domain recommendations without requiring direct access to domain-specific user interaction data, thereby preserving user privacy.

6.2 Model Merging

Model merging [4, 5, 8–11, 18, 21, 25, 31, 32, 38, 40, 43–46, 49, 50] aims to improve the generalization of domain-specific fine-tuned models by consolidating knowledge from multiple models trained on diverse domains or tasks into a single model, typically without requiring access to training data. The simplest approaches [10, 40] perform parameter averaging across fine-tuned models that share the same backbone pre-trained model.

Beyond these early methods, *task vector*-based model merging [4, 8, 9, 11, 25, 31, 38, 44, 46, 49, 50] has been proposed to enable more effective knowledge consolidation. Task vectors are defined as the parameter differences between each fine-tuned model and the pre-trained model. They can be interpreted as directions that encode domain- or task-specific adaptations. Task Arithmetic [9] shows that task vectors can be combined to build multi-task models, or negated to attenuate (or remove) task-specific knowledge. TIES [44] selects parameters with large task-induced changes and resolves sign conflicts to reduce interference. AdaMerging [46] extends linear task-vector composition by learning domain- or layer-wise merging weights through entropy minimization on unlabeled test samples. Although model merging has demonstrated strong effectiveness in computer vision and natural language processing, its potential for recommender systems has remained largely unexplored.

7 Conclusion

In this work, we addressed the fundamental limitations of existing CDSR under realistic constraints, where user interaction data cannot be shared across domains. To this end, we introduced ***MergeRec***, a novel framework that applies task vector-based model merging to a new problem setting, termed *data-isolated CDSR*. MergeRec consists of three key components: (1) *Merging initialization* constructs an initial merged model using training-free task vectors based merging. (2) *Pseudo-user data construction* synthesizes virtual interaction sequences from domain items, allowing CF signals to be extracted without exposing sensitive user data. (3) *Collaborative merging optimization* jointly optimizes a recommendation loss and a knowledge distillation loss, facilitating the transfer of domain-specific CF patterns while preserving ranking effectiveness. Extensive experiments confirmed that MergeRec consistently outperforms existing model merging baselines and significantly improves generalization, including on unseen and data-scarce domains, highlighting the potential of model merging as a scalable, privacy-preserving solution for building universal recommender systems.

Ethical Use of Data

This paper utilizes publicly available datasets (Amazon product review datasets) that contain no personally identifiable information and require no Institutional Ethics Review Board approval. All experiments use anonymized interaction sequences released for academic research purposes.

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References

[1] Jiangxia Cao, Xin Cong, Jiawei Sheng, Tingwen Liu, and Bin Wang. 2022. Contrastive Cross-Domain Sequential Recommendation. In *CIKM*. 138–147.

[2] Yukuo Cen, Jianwei Zhang, Xu Zou, Chang Zhou, Hongxia Yang, and Jie Tang. 2020. Controllable Multi-Interest Framework for Recommendation. In *KDD*. 2942–2951.

[3] Shu Chen, Zitao Xu, Weike Pan, Qiang Yang, and Zhong Ming. 2024. A Survey on Cross-Domain Sequential Recommendation. In *IJCAI*. 7989–7998.

[4] Guodong Du, Junlin Lee, Jing Li, Runhua Jiang, Yifei Guo, Shuyang Yu, Hanting Liu, Sim Kuan Goh, Ho-Kin Tang, Daojing He, and Min Zhang. 2024. Parameter Competition Balancing for Model Merging. In *NeurIPS*.

[5] Antonio Andrea Gargiulo, Donato Crisostomi, Maria Sofia Bucarelli, Simone Scardapane, Fabrizio Silvestri, and Emanuele Rodolà. 2025. Task Singular Vectors: Reducing Task Interference in Model Merging. In *CVPR*. 18695–18705.

[6] Balázs Hidasi, Alexandros Karatzoglou, Linas Baltrunas, and Domonkos Tikk. 2016. Session-based Recommendations with Recurrent Neural Networks. In *ICLR*.

[7] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. 2024. Bridging Language and Items for Retrieval and Recommendation. *arXiv preprint arXiv:2403.03952* (2024).

[8] Chenyu Huang, Peng Ye, Tao Chen, Tong He, Xiangyu Yue, and Wanli Ouyang. 2024. EMR-Merging: Tuning-Free High-Performance Model Merging. In *NeurIPS*.

[9] Gabriel Ilharco, Marco Túlio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. 2023. Editing models with task arithmetic. In *ICLR*.

[10] Gabriel Ilharco, Mitchell Wortsman, Samir Yitzhak Gadre, Shuran Song, Hannaneh Hajishirzi, Simon Kornblith, Ali Farhadi, and Ludwig Schmidt. 2022. Patching open-vocabulary models by interpolating weights. In *NeurIPS*.

[11] Ruochen Jin, Bojian Hou, Jiancong Xiao, Weijie J. Su, and Li Shen. 2025. Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic. In *ICLR*.

[12] Wang-Cheng Kang and Julian J. McAuley. 2018. Self-Attentive Sequential Recommendation. In *ICDM*. 197–206.

[13] Jiacheng Li, Ming Wang, Jin Li, Jinmiao Fu, Xin Shen, Jingbo Shang, and Julian J. McAuley. 2023. Text Is All You Need: Learning Language Representations for Sequential Recommendation. In *KDD*. 1258–1267.

[14] Muyang Li, Zijian Zhang, Xiangyu Zhao, Wanyu Wang, Minghao Zhao, Runze Wu, and Ruocheng Guo. 2023. AutoMLP: Automated MLP for Sequential Recommendations. In *WWW*. ACM, 1190–1198.

[15] Muyang Li, Xiangyu Zhao, Chuan Lyu, Minghao Zhao, Runze Wu, and Ruocheng Guo. 2022. MLP4Rec: A Pure MLP Architecture for Sequential Recommendations. In *IJCAI*. ijcai.org, 2138–2144.

[16] Guanyu Lin, Chen Gao, Yu Zheng, Jianxin Chang, Yanan Niu, Yang Song, Kun Gai, Zhiheng Li, Depeng Jin, Yong Li, and Meng Wang. 2024. Mixed Attention Network for Cross-domain Sequential Recommendation. In *WSDM*. 405–413.

[17] Qidong Liu, Xiangyu Zhao, Yejing Wang, Zijian Zhang, Howard Zhong, Chong Chen, Xiang Li, Wei Huang, and Feng Tian. 2025. Bridge the Domains: Large Language Models Enhanced Cross-domain Sequential Recommendation. In *SIGIR*. 1582–1592.

[18] Zhenyi Lu, Chenghao Fan, Wei Wei, Xiaoye Qu, Dangyang Chen, and Yu Cheng. 2024. Twin-Merging: Dynamic Integration of Modular Expertise in Model Merging. In *NeurIPS 2024*.

[19] Ziang Lu, Lei Guo, Xu Yu, Zhiyong Cheng, Xiaohui Han, and Lei Zhu. 2025. Federated Semantic Learning for Privacy-preserving Cross-domain Recommendation. *CoRR* abs/2503.23026 (2025).

[20] Muyang Ma, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Lifan Zhao, Peiyu Liu, Jun Ma, and Maarten de Rijke. 2022. Mixed Information Flow for Cross-Domain Sequential Recommendations. *ACM Trans. Knowl. Discov. Data* 16, 4 (2022), 64:1–64:32.

[21] Michael Matena and Colin Raffel. 2022. Merging Models with Fisher-Weighted Averaging. In *NeurIPS*.

[22] Jaewan Moon, Yoonki Jeong, Dong-Kyu Chae, Jaeho Choi, Hyunjung Shim, and Jongwuk Lee. 2023. CoMix: Collaborative filtering with mixup for implicit datasets. *Inf. Sci.* 628 (2023), 254–268.

[23] Jaewan Moon, Hye-young Kim, and Jongwuk Lee. 2023. It's Enough: Relaxing Diagonal Constraints in Linear Autoencoders for Recommendation. In *SIGIR*. 1639–1648.

[24] Jaewan Moon, Seongmin Park, and Jongwuk Lee. 2025. LLM-Enhanced Linear Autoencoders for Recommendation. In *CIKM*. 5036–5040.

[25] Guillermo Ortiz-Jiménez, Alessandro Favero, and Pascal Frossard. 2023. Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models. In *NeurIPS 2023*.

[26] Chung Park, Taesan Kim, Taekyoon Choi, Junui Hong, Yelim Yu, Mincheol Cho, Kyunam Lee, Sungil Ryu, Hyungjun Yoon, Minsung Choi, and Jaegul Choo. 2023. Cracking the Code of Negative Transfer: A Cooperative Game Theoretic Approach for Cross-Domain Sequential Recommendation. In *CIKM*. 2024–2033.

[27] Chung Park, Taesan Kim, Hyungjun Yoon, Junui Hong, Yelim Yu, Mincheol Cho, Minsung Choi, and Jaegul Choo. 2024. Pacer and Runner: Cooperative Learning Framework between Single- and Cross-Domain Sequential Recommendation. In *SIGIR*. 2071–2080.

[28] Seongmin Park, Mincheol Yoon, Jae-woong Lee, Hogun Park, and Jongwuk Lee. 2023. Toward a Better Understanding of Loss Functions for Collaborative Filtering. In *CIKM*. 2034–2043.

[29] Seongmin Park, Mincheol Yoon, Hye young Kim, and Jongwuk Lee. 2025. Why is Normalization Necessary for Linear Recommenders?. In *SIGIR*. 2142–2151.

[30] Anna Rogers, Olga Kovaleva, and Anna Rumshisky. 2020. A Primer in BERTology: What We Know About How BERT Works. *Trans. Assoc. Comput. Linguistics* 8 (2020), 842–866.

[31] Daiki Shirafuji, Makoto Takenaka, and Shinya Taguchi. 2025. Bias Vector: Mitigating Biases in Language Models with Task Arithmetic Approach. In *COLING*. 2799–2813.

[32] George Stoica, Daniel Bolya, Jakob Bjorner, Pratik Ramesh, Taylor Hearn, and Judy Hoffman. 2024. ZipIt! Merging Models from Different Tasks without Training. In *ICLR*.

[33] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *CIKM*. 1441–1450.

[34] Changxin Tian, Yuexiang Xie, Xu Chen, Yaliang Li, and Xin Zhao. 2024. Privacy-preserving Cross-domain Recommendation with Federated Graph Learning. *ACM Trans. Inf. Syst.* 42, 5 (2024), 135:1–135:29. https://doi.org/10.1145/3653448

[35] Yu Tian, Jianxin Chang, Yanan Niu, Yang Song, and Chenliang Li. 2022. When Multi-Level Meets Multi-Interest: A Multi-Grained Neural Model for Sequential Recommendation. In *SIGIR*. 1632–1641.

[36] Betty van Aken, Benjamin Winter, Alexander Löser, and Felix A. Gers. 2019. How Does BERT Answer Questions?: A Layer-Wise Analysis of Transformer Representations. In *CIKM*. 1823–1832.

[37] Hao Wang, Mingjia Yin, Luankang Zhang, Sirui Zhao, and Enhong Chen. 2025. MF-GSLAE: A Multi-Factor User Representation Pre-Training Framework for Dual-Target Cross-Domain Recommendation. *ACM Trans. Inf. Syst.* 43, 2 (2025), 30:1–30:28.

[38] Ke Wang, Nikolaos Dimitriadis, Guillermo Ortiz-Jiménez, François Fleuret, and Pascal Frossard. 2024. Localizing Task Information for Improved Model Merging and Compression. In *ICML*.

[39] Li Wang, Lei Sang, Quangui Zhang, Qiang Wu, and Min Xu. 2024. A privacy-preserving framework with multi-modal data for cross-domain recommendation. *Knowl. Based Syst.* 304 (2024), 112529.

[40] Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, and Ludwig Schmidt. 2022. Robust fine-tuning of zero-shot models. In *CVPR*. 7949–7961.

[41] Chuhan Wu, Fangzhao Wu, Yang Cao, Yongfeng Huang, and Xing Xie. 2021. FedGNN: Federated Graph Neural Network for Privacy-Preserving Recommendation. *CoRR* abs/2102.04925 (2021).

[42] Meihan Wu, Li Li, Chang Tao, Eric Rigall, Xiaodong Wang, and Cheng-Zhong Xu. 2022. FedCDR: Federated Cross-Domain Recommendation for Privacy-Preserving Rating Prediction. In *CIKM*. 2179–2188.

[43] Zhengqi Xu, Ke Yuan, Huiqiong Wang, Yong Wang, Mingli Song, and Jie Song. 2024. Training-Free Pretrained Model Merging. In *CVPR*. 5915–5925.

[44] Prateek Yadav, Derek Tam, Leshem Choshen, Colin A. Raffel, and Mohit Bansal. 2023. TIES-Merging: Resolving Interference When Merging Models. In *NeurIPS*.

[45] Enneng Yang, Li Shen, Zhenyi Wang, Guibing Guo, Xiaojun Chen, Xingwei Wang, and Dacheng Tao. 2024. Representation Surgery for Multi-Task Model Merging. In *ICML*.

[46] Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. 2024. AdaMerging: Adaptive Model Merging for Multi-Task Learning. In *ICLR*.

[47] Liu Yang, Ben Tan, Vincent W. Zheng, Kai Chen, and Qiang Yang. 2020. Federated Recommendation Systems. In *Federated Learning - Privacy and Incentive*. Vol. 12500. Springer, 225–239.

[48] Ziqi Yang, Zhaopeng Peng, Zihui Wang, Jianzhong Qi, Chaochao Chen, Weike Pan, Chenglu Wen, Cheng Wang, and Xiaoliang Fan. 2024. Federated Graph Learning for Cross-Domain Recommendation. In *NeurIPS*.

[49] Kotaro Yoshida, Yuji Naraki, Takafumi Horie, Ryosuke Yamaki, Ryotaro Shimizu, Yuki Saito, Julian J. McAuley, and Hiroki Naganuma. 2025. Mastering Task Arithmetic: τJp as a Key Indicator for Weight Disentanglement. In *ICLR*.

[50] Frederic Z. Zhang, Paul Albert, Cristian Rodriguez Opazo, Anton van den Hengel, and Ehsan Abbasnejad. 2024. Knowledge Composition using Task Vectors with Learned Anisotropic Scaling. In *NeurIPS 2024*.

[51] Shengyu Zhang, Lingxiao Yang, Dong Yao, Yujie Lu, Fuli Feng, Zhou Zhao, Tat-Seng Chua, and Fei Wu. 2022. Re4: Learning to Re-contrast, Re-attend, Re-construct for Multi-interest Recommendation. In *WWW*. 2216–2226.

[52] Feng Zhu, Chaochao Chen, Yan Wang, Guanfeng Liu, and Xiaolin Zheng. 2019. DTCDR: A Framework for Dual-Target Cross-Domain Recommendation. In *CIKM*. 1533–1542.

Table 5: Dataset statistics including the number of users, items, interactions, and density.

Dataset	# Users	# Items	# Inter.	Density
Arts	56,210	22,855	492,492	0.04%
Beauty	22,363	12,101	198,502	0.07%
Inst.	27,530	10,611	231,312	0.08%
Office	101,499	27,932	798,912	0.03%
Pantry	14,180	4,968	137,769	0.20%
Sci.	11,041	5,327	76,896	0.13%
Sports	35,598	18,357	296,337	0.05%
Toys	19,412	11,924	167,597	0.07%

A Dataset Statistics

Table 5 summarizes the statistics of each domain, including the number of users, items, interactions, and dataset density. The density is calculated as $\frac{\#\text{ Interactions}}{\#\text{ Users}\times\#\text{ Items}}$.

B Implementation Details

All methods, including MergeRec and baselines, are implemented in PyTorch. For RecFormer-base [13], we use the official pre-trained checkpoint[3], while for RecFormer-large, we pre-train the model following the protocol described in the original paper. For BLaIR-base[4] and BLaIR-large[5] [7], we use the official pre-trained checkpoints available on HuggingFace. Fine-tuning is performed with in-batch negative sampling and a batch size of 64. For merging baselines, we adopt their hyperparameter configurations for the validation-less setting, *i.e.*, no training, validation, or test data are used. Specifically, we set $w_1 = w_2 = \cdots = w_K = 0.4$ for Task Arithmetic [9], and $w = 1$ for TIES [44]. We use the top 20% of the parameters for TIES. For AdaMerging [46] and MergeRec, all coefficients are initialized to 0.2 and optimized for 500 steps using the Adam optimizer with a learning rate of 0.001 and a batch size of 16. For MergeRec, we set $\lambda = 1{,}000$ to balance the scale of the two loss functions. All reported performance metrics represent averages computed across five random seeds. For the significance test, we assume that deterministic merging approaches (*i.e.*, Weight Averaging, Task Arithmetic, and TIES) have identical performance values across all five runs.

C Overall Performance on Other Metrics

We report the unnormalized R@10 results in Table 6. **(i)** MergeRec achieves the best performance on average across all datasets and backbone models. This shows that MergeRec effectively transfers domain-specific knowledge and improves model discrimination, whereas AdaMerging is less effective across the board. **(ii)** Merging methods like MergeRec, Weight Averaging, and TIES consistently outperform joint learning, demonstrating that parameter merging can efficiently transfer knowledge across domains and reduce computational cost without cross-domain training data. **(iii)** Cross-domain merging is especially helpful for domains with limited data, such as Scientific, where MergeRec shows notable improvements over fine-tuned models due to more effective knowledge transfer.

[3]https://github.com/AaronHeee/RecFormer
[4]https://huggingface.co/hyp1231/blair-roberta-base
[5]https://huggingface.co/hyp1231/blair-roberta-large

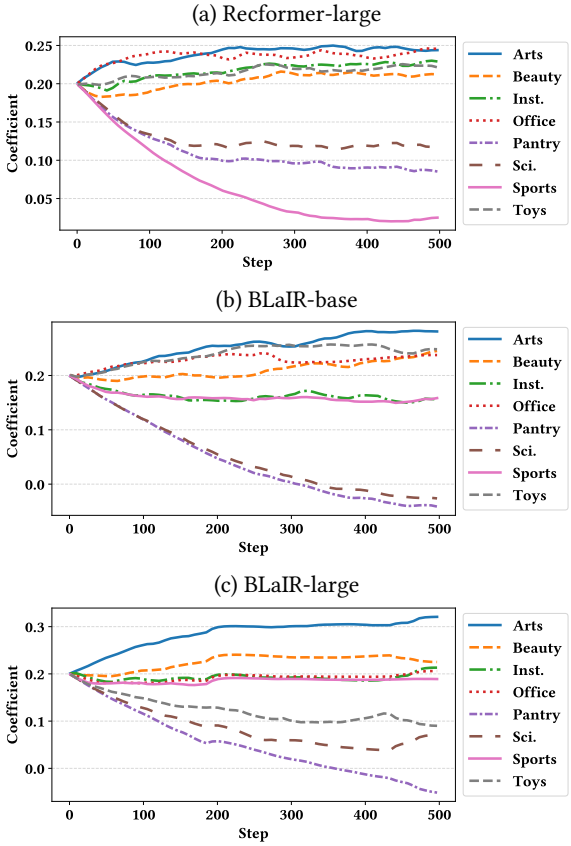


Figure 8: Domain-specific coefficients dynamics over training steps on RecFormer-large and BLaIR-base/large backbones.

D Domain-Specific Weight Dynamics on Other Backbones

We further analyze the evolution of domain-specific weights $\mathbf{w}$ on the remaining three backbone models: RecFormer-large and BLaIR-base/large. Consistent with our observations for RecFormer-base, we find that domains with larger data scales (*e.g.*, *Arts* and *Office*) tend to converge to higher weight coefficients. This trend suggests that these models also prioritize capturing more complex collaborative patterns in large-scale domains, thereby allocating more representational capacity to preserve domain-specific knowledge. In contrast to this general trend, the merging weight for the Pantry dataset in the BLaIR model converges to negative values. This behavior may be attributed to the absence of category information in the Pantry dataset, which differs from other domains and can lead to misalignment during model merging.

Table 6: Performance comparison with seven baseline methods on four backbone models, *i.e.*, RecFormer-base/large [13] and BLaIR-base/large [7]. We report absolute Recall@10 performance. The best results, excluding the fine-tuned model, are marked in bold, and the second-best results are shown as underlined. '*' indicates the statistically significant gain of MergeRec over the best baseline model (p < 0.02 for one-tailed t-test).

Backbone	Method	Avg.	Arts	Beauty	Inst.	Office	Pantry	Sci.	Sports	Toys
RecFormer-base	Zero-shot	0.0767	0.1192	0.0445	0.0717	0.0850	0.0664	0.1309	0.0255	0.0706
	Fine-tune	0.1017	0.1567	0.0699	0.0994	0.1379	0.0895	0.1409	0.0358	0.0831
	Joint Learning	0.0815	0.1212	0.0556	0.0828	0.1012	0.0764	0.1127	0.0249	0.0773
	Weight Averaging	0.0913	0.1429	0.0580	0.0903	0.1067	0.0838	0.1405	**0.0310**	0.0773
	Task Arithmetic	0.0904	0.1434	0.0569	0.0876	0.1156	0.0829	0.1394	0.0291	0.0685
	TIES	0.0926	0.1462	**0.0598**	0.0900	**0.1217**	0.0830	0.1377	0.0306	0.0717
	AdaMerging (Domain-wise)	0.0802	0.1367	0.0410	0.0754	0.0893	0.0671	0.1305	0.0243	0.0774
	AdaMerging (Layer-wise)	0.0685	0.1053	0.0230	0.0670	0.0807	0.0588	0.1192	0.0233	0.0706
	MergeRec (Domain-wise)	**0.0939***	**0.1507***	0.0585	0.0904	0.1162	**0.0855***	0.1420*	0.0301	**0.0776***
	MergeRec (Layer-wise)	0.0936*	0.1496*	0.0593	**0.0906***	0.1145	0.0849*	**0.1430***	0.0304	0.0766
RecFormer-large	Zero-shot	0.0613	0.1138	0.0374	0.0470	0.0545	0.0415	0.1200	0.0205	0.0557
	Fine-tune	0.1038	0.1568	0.0721	0.1021	0.1388	0.0935	0.1463	0.0327	0.0877
	Joint Learning	0.0869	0.1304	0.0573	0.0815	0.1022	0.0844	0.1360	0.0299	0.0733
	Weight Averaging	0.0947	0.1447	0.0628	0.0908	0.1098	**0.0902**	**0.1439**	**0.0323**	0.0827
	Task Arithmetic	0.0913	0.1432	0.0601	0.0901	0.1149	0.0861	0.1360	0.0287	0.0711
	TIES	0.0933	0.1458	0.0618	0.0922	**0.1224**	0.0869	0.1377	0.0291	0.0708
	AdaMerging (Domain-wise)	0.0753	0.1307	0.0429	0.0792	0.0885	0.0574	0.1228	0.0190	0.0620
	AdaMerging (Layer-wise)	0.0735	0.1248	0.0377	0.0697	0.0842	0.0628	0.1216	0.0239	0.0631
	MergeRec (Domain-wise)	**0.0965***	**0.1492***	**0.0650***	**0.0937***	0.1157	0.0900	0.1425	0.0310	**0.0847***
	MergeRec (Layer-wise)	0.0960*	0.1477*	0.0647*	0.0922	0.1141	0.0900	0.1432	0.0316	0.0843*
BLaIR-base	Zero-shot	0.0409	0.0704	0.0224	0.0309	0.0376	0.0431	0.0759	0.0110	0.0355
	Fine-tune	0.0995	0.1535	0.0631	0.0980	0.1356	0.0913	0.1382	0.0320	0.0840
	Joint Learning	0.0832	0.1272	**0.0614**	**0.0819**	0.0991	0.0833	0.1034	**0.0297**	**0.0797**
	Weight Averaging	0.0874	0.1412	0.0593	0.0768	0.1032	**0.0834**	**0.1381**	0.0286	0.0687
	Task Arithmetic	0.0612	0.0876	0.0337	0.0653	0.0835	0.0570	0.1013	0.0158	0.0452
	TIES	0.0825	0.1295	0.0492	0.0767	**0.1188**	0.0705	0.1223	0.0279	0.0650
	AdaMerging (Domain-wise)	0.0603	0.1043	0.0329	0.0523	0.0608	0.0464	0.1008	0.0233	0.0614
	AdaMerging (Layer-wise)	0.0681	0.1122	0.0386	0.0675	0.0750	0.0525	0.1160	0.0238	0.0593
	MergeRec (Domain-wise)	0.0869	**0.1449***	0.0564	0.0762	0.1104	0.0778	0.1320	0.0272	0.0706
	MergeRec (Layer-wise)	**0.0875**	0.1436*	0.0579	0.0774	0.1100	0.0802	0.1334	0.0280	0.0698
BLaIR-large	Zero-shot	0.0333	0.0589	0.0205	0.0246	0.0257	0.0429	0.0501	0.0096	0.0339
	Fine-tune	0.0995	0.1534	0.0683	0.1003	0.1334	0.0923	0.1325	0.0346	0.0813
	Joint Learning	0.0842	0.1278	0.0553	0.0871	0.1048	0.0836	0.1040	0.0290	**0.0818**
	Weight Averaging	0.0886	0.1423	0.0609	0.0841	0.1017	0.0860	0.1372	0.0299	0.0664
	Task Arithmetic	0.0785	0.1268	0.0540	0.0661	0.0993	0.0772	0.1240	0.0253	0.0552
	TIES	0.0905	0.1439	0.0622	0.0881	**0.1201**	0.0854	0.1303	**0.0324**	0.0613
	AdaMerging (Domain-wise)	0.0689	0.1261	0.0426	0.0720	0.0683	0.0562	0.1039	0.0267	0.0556
	AdaMerging (Layer-wise)	0.0778	0.1301	0.0504	0.0788	0.0833	0.0766	0.1114	0.0247	0.0675
	MergeRec (Domain-wise)	0.0913*	0.1500*	0.0636*	**0.0921***	0.1074	0.0828	0.1363	0.0312	0.0673
	MergeRec (Layer-wise)	**0.0932***	**0.1515***	**0.0651***	0.0921*	0.1102	**0.0869***	**0.1394***	0.0312	0.0696